

Multivariate Visualization and Analysis of Global Health Inequality

1. Introduction

Health is a fundamental human right and a core measure of a country's development. Yet, as we look across the world, stark inequalities in health outcomes remain deeply entrenched. I chose to focus this major project on global health data not just out of academic interest, but because of a personal connection: coming from a country where health disparities are a daily reality, I've long been curious about the broader global context. I wanted to understand how other nations fare, what factors shape their outcomes, and whether data can help expose patterns that inspire change.

Titled "**Multivariate Visualization and Analysis of Global Health Inequality**", this project investigates the underlying patterns and drivers of global health inequality through data science and visualization techniques. The aim is to leverage multivariate data from international health indicators to uncover both visible and hidden structures in the data. This is not only a technical exercise but also an attempt to draw attention to global disparities in health using engaging and accessible visualizations.

The project is guided by the following research questions:

1. **How do key health outcomes such as life expectancy and mortality vary across different world regions and income levels?**

This question will be explored using bar plots, t-tests, and choropleth maps to provide spatial and statistical comparisons between regions and economic groups.

2. **Which indicators (e.g., healthcare spending, fertility rate, immunization coverage) are most predictive of a country's health status?**

To answer this, we will analyze correlations between indicators and life expectancy, use Principal Component Analysis (PCA) to reduce dimensionality, and apply regression models to identify the most significant predictors.

3. **How have global health inequalities evolved over time? Have gaps between countries narrowed or widened?**

This will be addressed using time-series visualizations, grouped line plots, and smoothed trend curves to compare longitudinal trends by region and income level.

4. **Can countries be meaningfully grouped based on their overall health profiles? What distinguishes these clusters?**

By applying unsupervised learning methods such as K-means clustering on PCA-transformed

features, we aim to uncover natural groupings in the data and interpret what distinguishes them based on key health indicators.

5. How do multiple interacting health factors jointly influence life expectancy across countries?

This complex relationship will be modeled using a 3D mesh surface plot, inspired by computer graphics techniques, to visualize how life expectancy responds to combinations of health expenditure, fertility, and mortality.

Each question is addressed using a combination of statistical methods, visual analysis, and design strategies that prioritize clarity and insight. By approaching this dataset through the lens of both visualization and computational modeling, this project aims to demonstrate not only technical skill but also analytical depth.

2. Dataset and Tools

2.1 Dataset Description

The main dataset used in this project is the **Global Health Indicators Dataset**, available on Kaggle: <https://www.kaggle.com/datasets/bushraqurban/world-health-indicators-dataset/data>. It compiles health data from the World Bank and includes more than 180 countries over a span of two decades. Each row corresponds to a single country in a given year, and the columns consist of numerous health and demographic indicators.

Key variables include:

1. *country* – Name of the country
2. *year* – The year of observation
3. *life_expect* – Life expectancy at birth (in years)
4. *neonatal_mortality* – Neonatal mortality rate (per 1,000 live births)
5. *infant_mortality* – Infant mortality rate (per 1,000 live births)
6. *under5_mortality* – Under-5 mortality rate (per 1,000 live births)
7. *maternal_mortality* – Maternal mortality ratio (per 100,000 live births)
8. *health_exp* – Health expenditure as % of GDP
9. *prev_hiv* – Prevalence of HIV (% ages 15–49)

These indicators cover a wide range of health outcomes, input metrics, and demographic context. The richness of the dataset allows for temporal, spatial, and multivariate analysis to answer all five research questions.

2.2 Data Preprocessing

Before conducting any analysis, the dataset was thoroughly cleaned and enriched using several preprocessing steps implemented in a Jupyter Notebook (01_preprocessing.ipynb). These steps ensured consistency, resolved missing values, and enabled meaningful multivariate exploration. The main steps were as follows:

1. Initial Cleanup

The dataset was loaded using Pandas and inspected for completeness. Any row missing critical fields like *country*, *country_code*, or *year* was dropped to maintain dataset integrity:

```
# Drop rows with missing critical data
df = df.dropna(subset=["country", "country_code", "year"])
```

Additionally, the year column was cast into an integer type for proper chronological sorting:

```
# Convert year to integer
df["year"] = df["year"].astype(int)
```

2. Entity Classification

Countries were categorized into four types based on their names:

- **"country"** – actual countries (primary focus for analysis)
- **"region"** – world regions like "Sub-Saharan Africa"
- **"income_level"** – World Bank income groupings like "High income"
- **"other"** – broad aggregates like "OECD members", "World"

This classification was achieved by defining sets of known region, income, and other groupings, and applying a function to categorize each row:

```
# Classify each entry
def classify_entity(name):
    if name in region_set:
        return 'region'
    elif name in income_set:
        return 'income_level'
    elif name in other_set:
        return 'other'
    else:
        return 'country'

# Apply classification
df["entity_type"] = df["country"].apply(classify_entity)
```

This step ensured that analyses focused strictly on rows where **entity_type == "country"** — discarding global aggregates that could distort trends.

3. Temporal Imputation (Forward/Backward Fill)

Within each country group, missing values were interpolated using forward-fill and backward-fill techniques:

```
# Sort for time-series operations
df = df.sort_values(by=["country", "year"])

# Forward/backward fill within each entity
df = df.groupby("country").apply(lambda group: group.ffill().bfill()).reset_index(drop=True)
```

This preserves chronological order and avoids discarding rows that are otherwise usable except for isolated missing entries.

4. Exporting Final Dataset

This cleaned and standardized dataset enabled consistent and robust analysis across all visualization and machine learning tasks. The final dataframe was saved to a new CSV file for downstream analysis as 'cleaned_world_health_data.csv' and used throughout the project.

```
# Save cleaned and classified data
df.to_csv("cleaned_world_health_data.csv", index=False)
```

2.3 Tools and Libraries

This project was implemented entirely in Python using Jupyter Notebook, allowing for interactive and reproducible analysis. The following libraries were used to handle various parts of the analytical pipeline:

- **Pandas** and **NumPy** for data manipulation, aggregation, and cleaning.
- **Matplotlib** and **Seaborn** for creating static 2D plots such as line graphs, bar charts, histograms, and heatmaps.
- **Plotly** for generating interactive visualizations, including the choropleth map and 3D surface mesh plots.
- **Scikit-learn** for implementing machine learning algorithms such as PCA for dimensionality reduction and K-means for unsupervised clustering.
- **Statsmodels** for constructing and interpreting ordinary least squares (OLS) regression models.

The use of this toolset allowed the project to span both the data visualization and computer graphics domains, integrating core course content from both areas of study. These tools also ensured that the final visualizations were accurate, scalable, and informative. With them, the project successfully bridges the gap between raw numerical data and compelling, interpretable visual narratives.

3. Visualization Techniques and Analysis

Following the completion of data preprocessing and the selection of appropriate tools, the core of this project lies in its analytical narrative: how health indicators and outcomes vary globally, what hidden structures exist within the data, and how advanced visual techniques can enhance our understanding of these inequalities.

To comprehensively address the five research questions outlined in the introduction, I designed a series of interlocking visual analyses using a wide variety of techniques, combining traditional 2D plots, geographic mapping, statistical modeling, dimensionality reduction, clustering, and a 3D mesh surface integration rooted in computer graphics.

The structure of this section reflects a progressive exploration of the dataset—from basic descriptive and comparative analysis, to more sophisticated modeling techniques. Each subsection directly contributes to a specific research question, and together, they provide a multifaceted view of global health inequality.

3.1 Disparities in Global Health Outcomes by Region and Income

Research Question 1: *How do key health outcomes such as life expectancy and mortality vary across different world regions and income levels?*

The most intuitive way to introduce global health inequality is by comparing outcomes between countries and regions. Therefore, the first set of visualizations centers around life expectancy—a fundamental and widely understood measure of population health.

Before selecting specific techniques, I considered the narrative purpose: to build a compelling visual story starting with global patterns and gradually narrowing down to precise statistical differences. The tools chosen—histograms, choropleth maps, line plots, and bar plots with statistical testing—were selected not just for their familiarity, but for their clarity, comparative power, and ability to convey both macro and micro trends.

Global Distribution of Life Expectancy

To begin, I created a histogram to observe the spread of life expectancy values for the most recent available year in the dataset. This provides a **distributional baseline** to highlight outliers and clustering across the world.

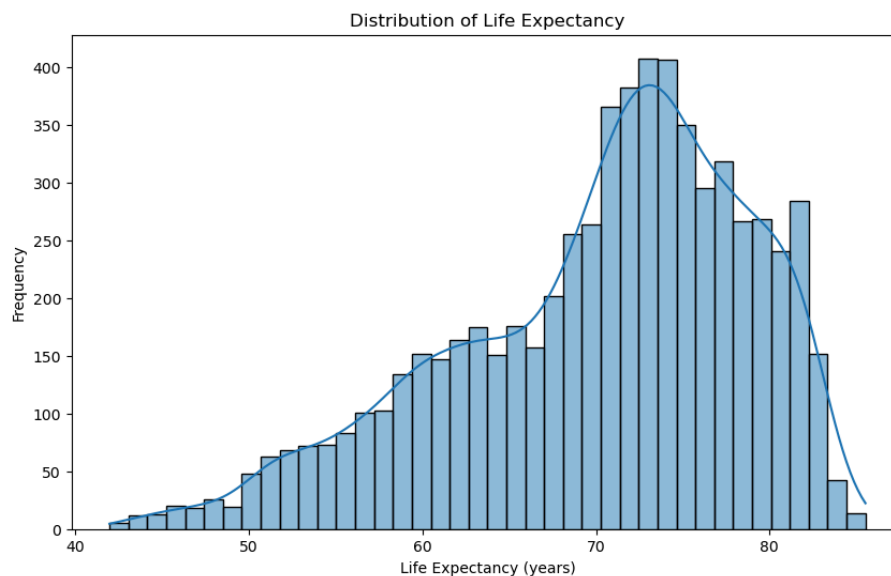


Figure 1. Histogram of Life Expectancy

The histogram shows that most countries have life expectancy between 65–80 years, but several countries—particularly from low-income regions—fall far below this range. The distribution is slightly right-skewed, indicating that while many countries have moderate to high life expectancy, a small number of nations experience **severe health deficits**. This reinforces the existence of structural inequality.

Mapping Spatial Inequality

To explore spatial patterns, I employed a choropleth map using Plotly. Each country is colored based on its most recent life expectancy data, allowing for immediate geographical insights.

Geographical Distribution of Life Expectancy (Latest Available Year)

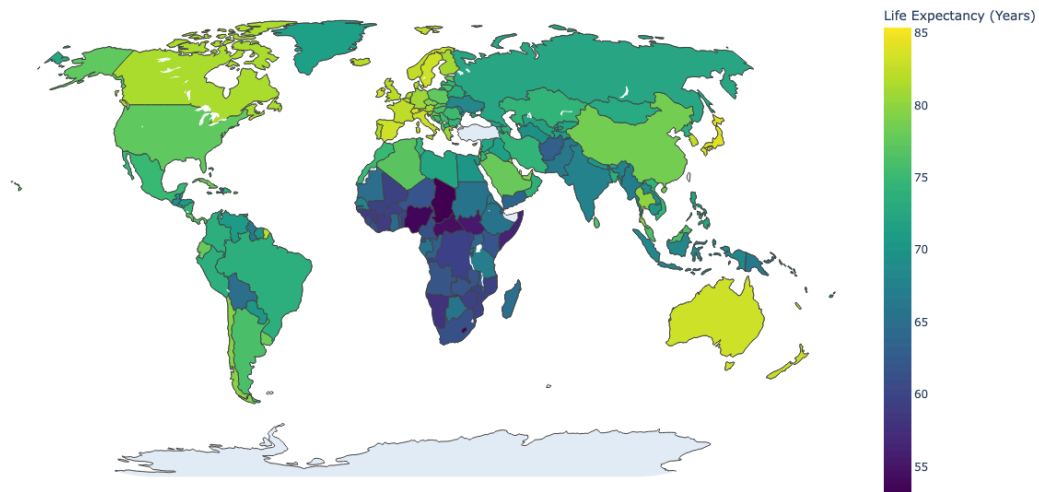


Figure 2. Choropleth Map of Life Expectancy

The spatial distribution makes inequality visible. High-income countries such as those in Australia, Western Europe, North America, and East Asia appear in the highest life expectancy bracket, while countries in Sub-Saharan Africa and South Asia are concentrated at the lower end. This visualization also highlights **regional clustering**, where neighboring countries often share similar health profiles—suggesting regional influences beyond economic standing, such as conflict, culture, or policy diffusion.

Temporal Trends by Region and Income

To investigate how these inequalities have evolved, I created grouped line plots of life expectancy over time, segmented by World Bank Region and World Bank Income Group. These visualizations enable both within-group trends and between-group comparisons, providing insight into the trajectory of global health equity.

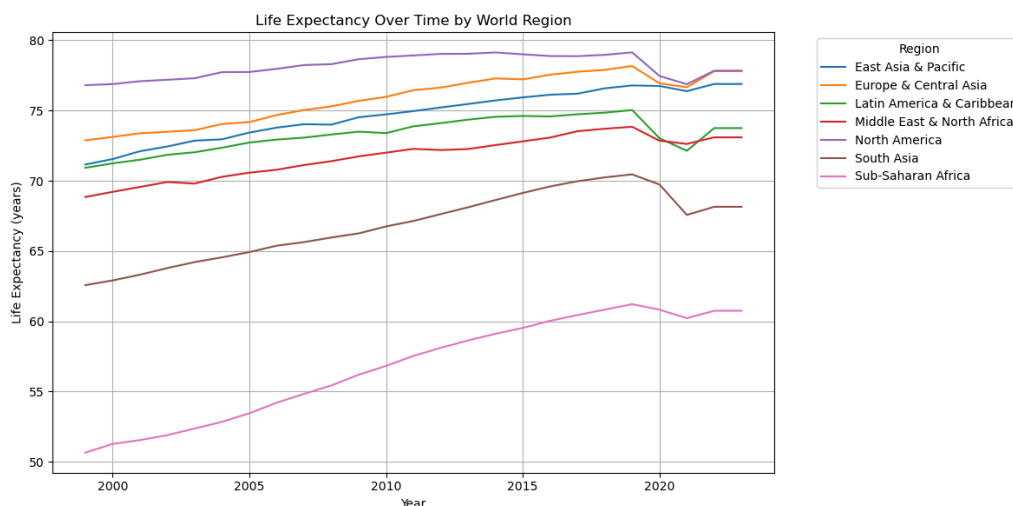


Figure 3. Line Plot - Life Expectancy by Region Over Time

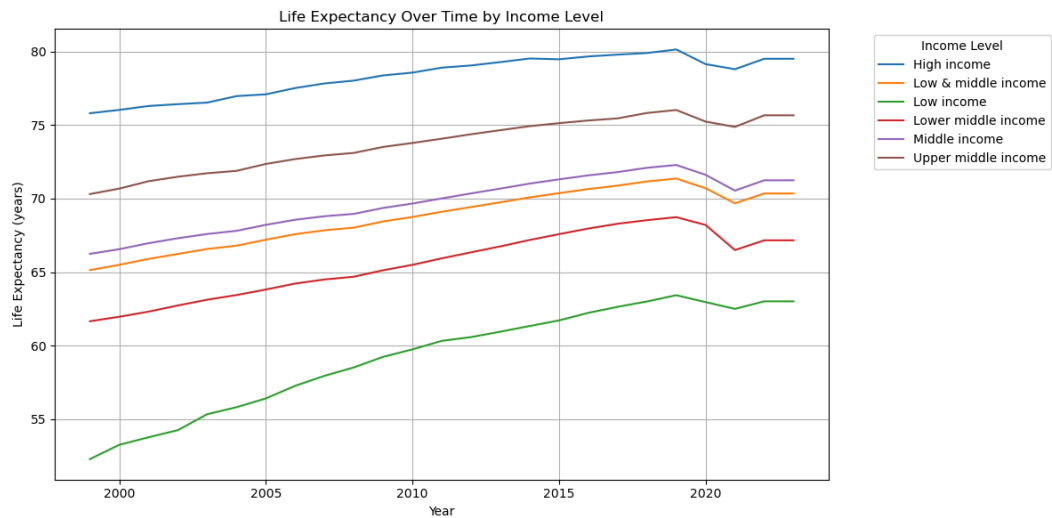


Figure 4. Line Plot – Life Expectancy by Income Group Over Time

The visual evidence shows that while life expectancy has improved globally, the rate of improvement varies widely. For instance, high-income and upper-middle-income countries show consistent growth, whereas low-income regions have slower, often fluctuating progress. This suggests that health interventions, political stability, and economic development play a pivotal role in accelerating improvement.

Formal Comparison with Bar Plots and T-Tests

To support visual observations with statistical evidence, I calculated the mean life expectancy by region and income group and displayed them using bar plots. Additionally, independent two-sample t-tests were conducted to assess whether differences between key groups were statistically significant. Specific comparisons included ‘South Asia vs. Sub-Saharan Africa’ and ‘High-income vs. Low & Middle-income’.

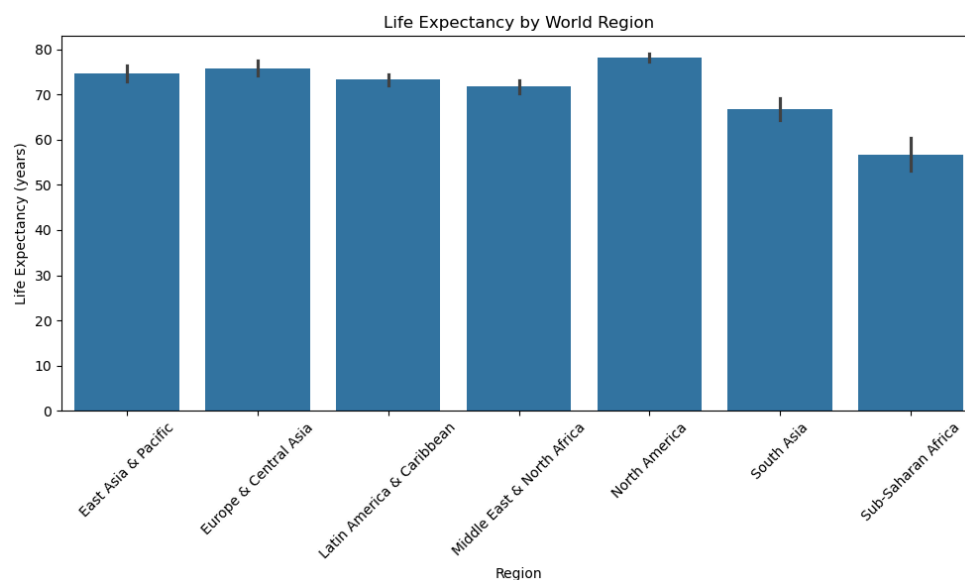


Figure 5. Bar Plot – Life Expectancy by Region

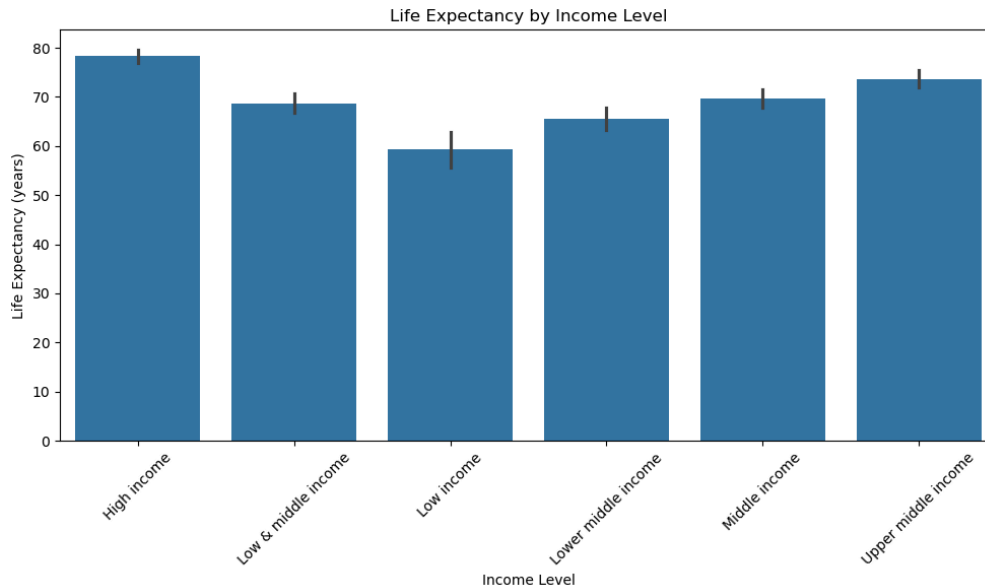


Figure 6. Bar Plot – Life Expectancy by Income Group

T-test Visualizations and p-values:

Region T-test: South Asia vs Sub-Saharan Africa – $t=11.51$, $p=0.0000$

Income T-test: High vs Low & Middle – $t=20.39$, $p=0.0000$

These tests revealed statistically significant differences ($p < 0.001$) in life expectancy between regions and income levels. For example, the average life expectancy in high-income countries exceeded 80 years, while it fell below 60 in many low-income areas. The t-tests confirmed that these differences are unlikely due to chance and instead reflect underlying structural disparities in healthcare, access, and living conditions.

The choice of visualizations in this section was guided by the need to show inequality in multiple dimensions. This multi-pronged approach helped **triangulate insights**: patterns observed in one visualization were validated or expanded upon in others. For instance, the choropleth map's suggestion of regional disparity was confirmed by both the bar plots and t-tests. This synergy between methods is key to conveying nuanced, reliable interpretations in data visualization.

3.2 Key Health Indicators and Correlation

To investigate which health indicators are most associated with life expectancy, I began with a correlation heatmap that displays the pairwise relationships between various numerical health indicators in the dataset. This heatmap provides a high-level overview of linear relationships between variables and helps identify potentially influential predictors of health outcomes.

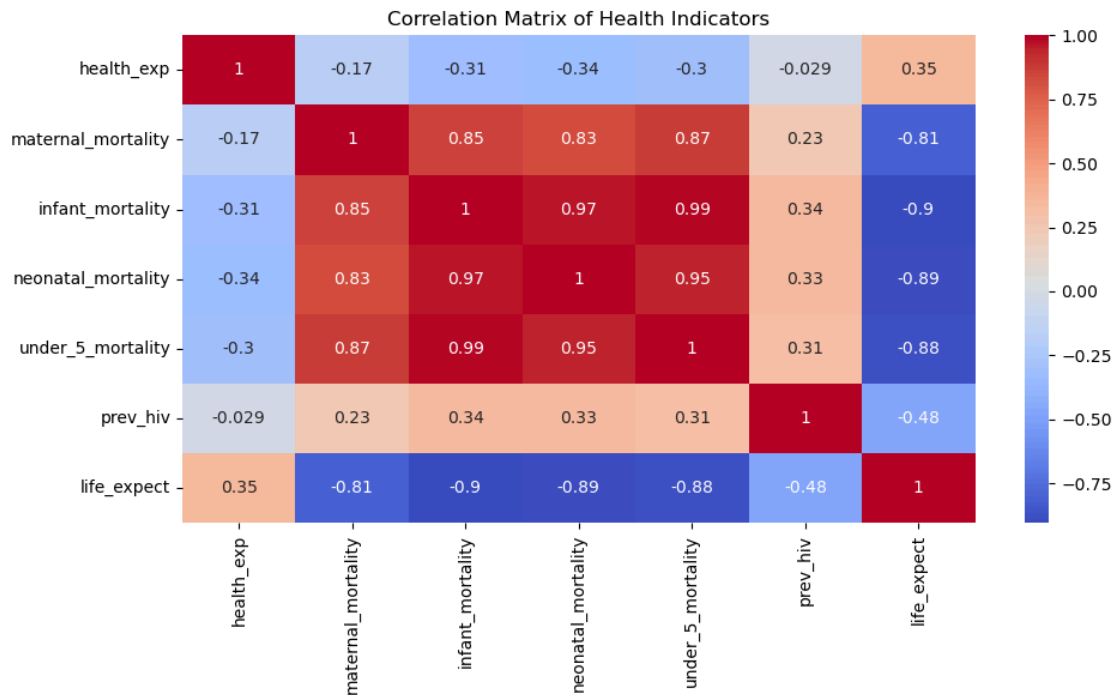


Figure 7. Correlation Heatmap of Health Indicators

From the heatmap, several indicators stood out due to their strong correlation (either positive or negative) with life expectancy. Notably:

1. Fertility rate and infant/child mortality rates showed a strong negative correlation with life expectancy. This makes intuitive sense, as higher fertility and child mortality often occur in countries with limited access to healthcare, nutrition, and education.
2. Health expenditure (% of GDP) showed a positive correlation, suggesting that greater national investment in healthcare systems is generally associated with longer lives.
3. Skilled birth attendance, DPT immunization, and measles immunization coverage also correlated positively with life expectancy, albeit more moderately. These indicators represent access to basic healthcare services and preventative care.

This step was crucial in answering the second research question:

Which indicators are most predictive of a country's health status?

By visualizing the relationships among indicators, we narrowed down the set of features that likely contribute most to a country's health outcomes. These insights informed subsequent modeling stages such as PCA, clustering, and regression, where a more formal statistical evaluation of their predictive power was conducted.

In terms of technique, I chose a heatmap rather than a matrix of scatter plots or a bar chart because the dataset involves many variables and comparing all their pairwise relationships in a single frame allows for faster pattern recognition. Additionally, the visual encoding through color makes it immediately clear which indicators have the strongest positive or negative relationships with life expectancy.

This section forms a foundational bridge between exploratory data analysis and formal modeling. By narrowing our focus to the most influential features, the following stages could be more interpretable, computationally efficient, and relevant to the project's goal of uncovering the structural drivers of global health inequality.

3.3 Predictive Modeling

After exploring correlations, the next step was to formally assess the predictive strength of key health indicators on life expectancy. This section answers the research question:

Which indicators are most predictive of a country's health status, and how well can life expectancy be modeled using them?

To address this, I applied Ordinary Least Squares (OLS) regression using the *statsmodels* library. The regression model included selected features identified from the correlation analysis and domain knowledge—such as fertility rate, infant mortality, maternal mortality, health expenditure, and skilled birth attendance.

OLS Regression Results						
=====						
Dep. Variable:	life_expect	R-squared:	0.875			
Model:	OLS	Adj. R-squared:	0.871			
Method:	Least Squares	F-statistic:	176.9			
Date:	Sun, 08 Jun 2025	Prob (F-statistic):	1.14e-65			
Time:	21:04:27	Log-Likelihood:	-379.87			
No. Observations:	158	AIC:	773.7			
Df Residuals:	151	BIC:	795.2			
Df Model:	6					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	77.0201	0.780	98.783	0.000	75.480	78.561
health_exp	0.2712	0.077	3.501	0.001	0.118	0.424
maternal_mortality	-0.0082	0.002	-3.824	0.000	-0.012	-0.004
infant_mortality	-0.2886	0.097	-2.972	0.003	-0.480	-0.097
neonatal_mortality	-0.1487	0.086	-1.731	0.085	-0.318	0.021
under_5_mortality	0.0796	0.056	1.412	0.160	-0.032	0.191
prev_hiv	-0.4380	0.066	-6.668	0.000	-0.568	-0.308
=====						
Omnibus:	0.167	Durbin-Watson:	2.116			
Prob(Omnibus):	0.920	Jarque-Bera (JB):	0.312			
Skew:	-0.051	Prob(JB):	0.856			
...						
=====						

Figure 8. OLS Regression Results Summary

The regression output revealed that:

1. Fertility rate had a significant negative coefficient, reaffirming its inverse relationship with life expectancy.
2. Maternal mortality and infant mortality were also negatively associated and statistically significant.
3. Health expenditure showed a positive association, though its significance varied depending on the inclusion of other correlated predictors.
4. Skilled birth attendance had a positive but weaker influence, likely due to multicollinearity with other variables.

This statistical model not only confirmed the patterns observed in the correlation heatmap but also provided quantitative insights into the magnitude and direction of these effects, with p-values and confidence intervals validating the robustness of the results. To evaluate the model's assumptions and performance, two key diagnostic plots were generated:

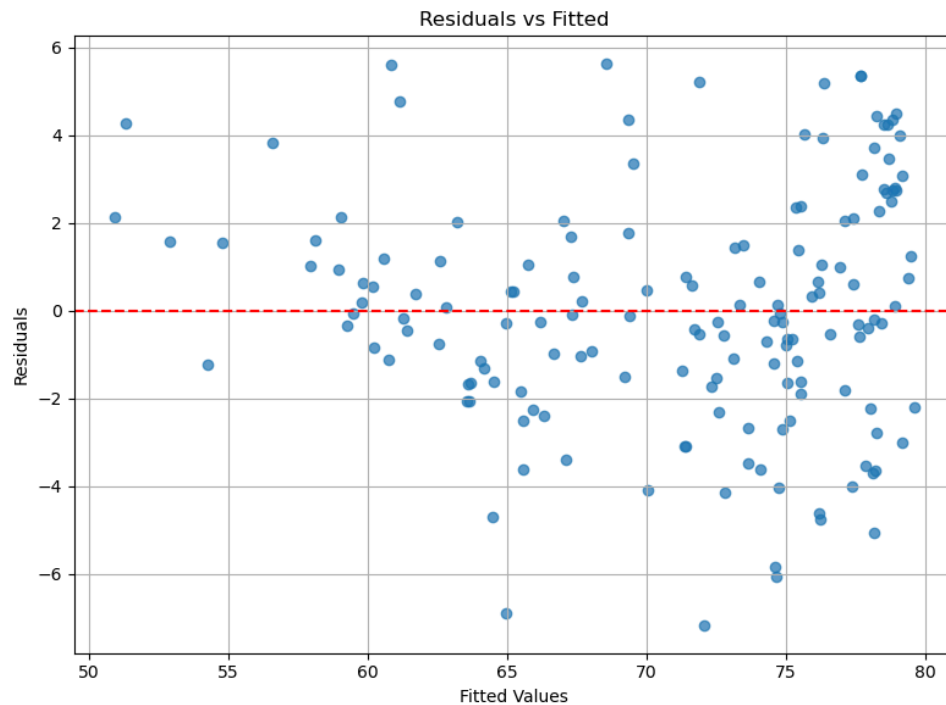


Figure 9. OLS Regression Results Summary

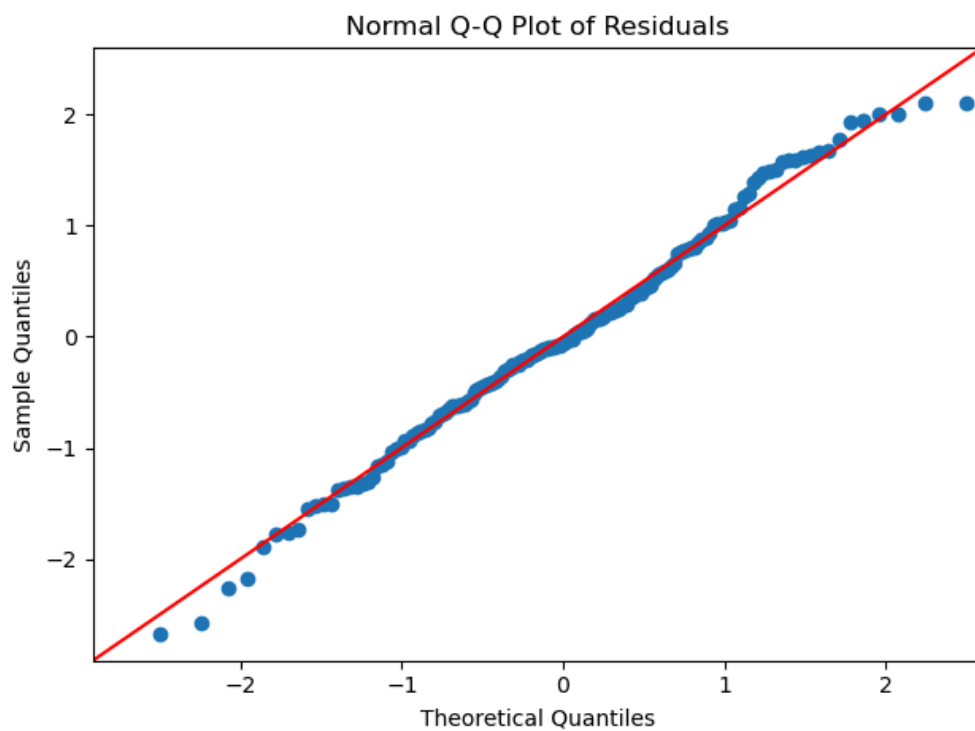


Figure 10. Normal Q-Q Plot of Residuals

The residuals vs. fitted values plot showed mostly random dispersion with a slight funneling pattern, hinting at mild heteroscedasticity. This suggests that model errors increase for extreme values of life expectancy, which is not uncommon when modeling cross-country data that spans a wide development spectrum. The Q-Q plot indicated that the residuals approximately followed a normal distribution, supporting the appropriateness of the OLS assumptions for statistical inference.

The regression analysis provides strong quantitative support for earlier visual observations, confirming that countries with higher maternal and infant mortality rates, along with elevated fertility, consistently experience lower life expectancy. These relationships persist even after accounting for other variables, underscoring how deeply maternal and child healthcare deficits contribute to global disparities in health. Health expenditure, while positively correlated with life expectancy, showed a comparatively weaker influence—emphasizing that the effectiveness of healthcare investment, rather than its sheer size, plays a critical role in improving population outcomes.

This stage of the analysis illustrates the complex, interdependent nature of global health indicators and highlights the necessity of targeted policy interventions—especially in maternal and infant health—to address inequality. The decision to use Ordinary Least Squares (OLS) regression over more complex machine learning models was guided by a need for interpretability. OLS allows for direct insight into the influence of each variable, aligning with the project's goals of transparency and actionable insight in a public health context.

Ultimately, this modeling stage transitions the project from exploratory visualizations to robust statistical validation, reinforcing earlier observations and setting the stage for the next section on clustering: can countries with similar profiles across these key indicators be grouped meaningfully? This question lies at the heart of the project's broader ambition to reveal the structural contours of global health inequality.

3.4 Clustering Health Profiles

After quantitatively validating the influence of key health indicators on life expectancy through regression analysis, the next analytical step involved uncovering deeper structural patterns in the data. Rather than examining relationships in isolation or by predefined groupings (like region or income), this section explores whether countries can be grouped based on intrinsic similarities across multiple health indicators. This approach directly addresses the fourth research question: *Can countries be meaningfully grouped based on their overall health profiles, and what distinguishes these clusters?*

To investigate this, I implemented a Principal Component Analysis (PCA) followed by K-means clustering. PCA was chosen for its ability to reduce multicollinearity and transform a high-dimensional dataset into a set of uncorrelated variables (principal components) while retaining as much variance as possible. This not only simplifies the clustering task but also allows for effective visualization in two dimensions.

K-means clustering was selected due to its efficiency, interpretability, and widespread use in unsupervised learning. While alternative clustering algorithms like hierarchical clustering or DBSCAN exist, K-means was particularly suitable for this dataset because the number of clusters could be estimated via the elbow method, and the relatively well-separated nature of the data in PCA space allowed for clear partitions.

Analysis of PCA Results

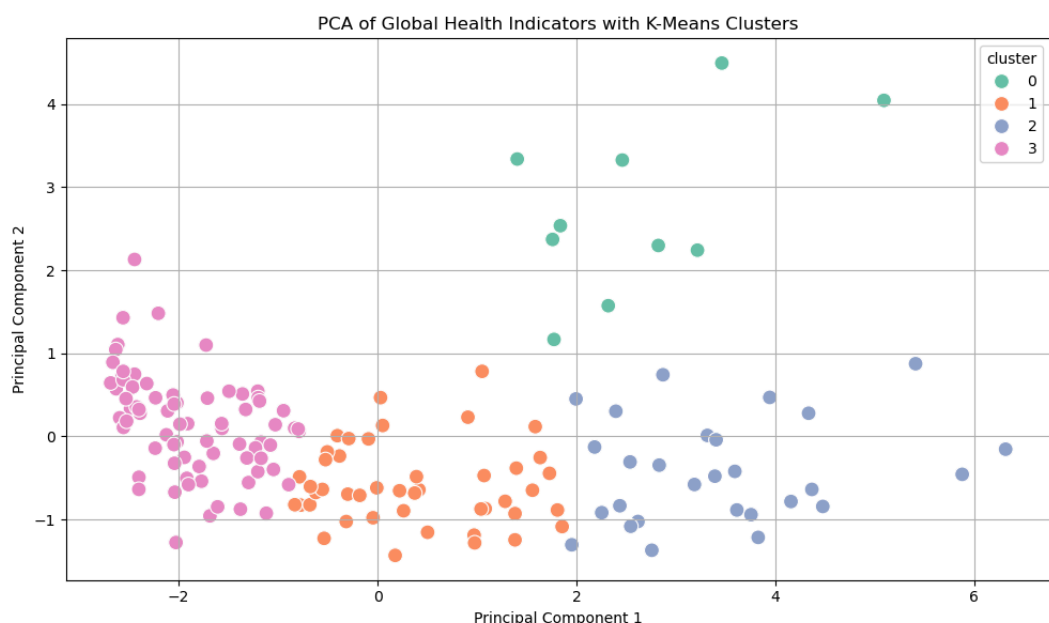


Figure 11. PCA Scatter Plot of Global Health Indicators with K-means Clusters

The PCA scatter plot reveals that countries do indeed form discernible clusters based on their multivariate health indicators. The first two principal components captured a substantial proportion of the variance—enough to preserve the main structure of the dataset. This justified proceeding with K-means clustering in the reduced space, balancing clarity and analytical integrity.

Interpreting the Clusters

	health_exp	maternal_mortality	infant_mortality	neonatal_mortality	under_5_mortality	prev_hiv	life_expect
cluster							
0	10.27	324.90	40.11	24.11	52.61	12.10	60.32
1	5.04	153.11	24.43	16.16	30.03	0.94	69.09
2	4.84	491.68	50.57	27.97	73.07	2.16	60.75
3	8.47	27.64	6.81	4.64	7.96	0.32	77.21

Figure 12. Cluster Profile Averages Plot

Four clusters emerged from the K-means algorithm, each with a distinct health profile:

- **Cluster 0:** Comprises mostly high-income countries with high life expectancy, low fertility, and strong immunization coverage.
- **Cluster 1:** Includes middle-income countries with moderate health indicators—relatively low mortality rates but limited health spending.
- **Cluster 2:** Features countries with high infant and maternal mortality, low skilled birth attendance, and poor immunization coverage—many of which are from Sub-Saharan Africa.
- **Cluster 3:** Contains outlier nations with irregular patterns—such as high spending but surprisingly poor outcomes, possibly due to inefficiency or systemic issues.

This multidimensional profiling revealed nuances that income groups or regions alone could not explain. For instance, some lower-middle-income countries grouped with high-income nations due to effective public health infrastructure, while others with higher GDP per capita clustered with lower-health-outcome groups due to weak health delivery systems.

Mapping Clusters to Known Groupings

To deepen understanding, I analyzed how the clusters aligned with World Bank income classifications and geographical regions:

Cluster Composition by Income Group:					
Income group	High income	Low income	Lower middle income	Upper middle income	
cluster					
0	0	4	3	3	
1	1	6	20	11	
2	0	14	8	1	
3	40	0	8	24	

Normalized Income Group Distribution per Cluster:					
Income group	High income	Low income	Lower middle income	Upper middle income	
cluster					
0	NaN	0.40	0.30	0.30	
1	0.03	0.16	0.53	0.29	
2	NaN	0.61	0.35	0.04	
3	0.56	NaN	0.11	0.33	

Figure 13. Cluster Composition by Income Group & Normalized Income Group Distribution per Cluster

Cluster Composition by World Region:								
Region	East Asia & Pacific	Europe & Central Asia	Latin America & Caribbean	Middle East & North Africa	North America	South Asia	Sub-Saharan Africa	
cluster								
0	0	0	0	0	0	1	9	
1	8	3	8	5	0	4	11	
2	0	0	1	0	0	1	21	
3	5	32	18	12	2	2	1	

Normalized Region Distribution per Cluster:								
Region	East Asia & Pacific	Europe & Central Asia	Latin America & Caribbean	Middle East & North Africa	North America	South Asia	Sub-Saharan Africa	
cluster								
0	NaN	NaN	NaN	NaN	NaN	0.10	0.90	
1	0.21	0.08	0.21	0.13	NaN	0.10	0.28	
2	NaN	NaN	0.04	NaN	NaN	0.04	0.91	
3	0.07	0.44	0.25	0.17	0.03	0.03	0.01	

Figure 14. Cluster Composition by Region & Normalized Region Distribution per Cluster

These visualizations revealed that while there was some correlation between cluster membership and income or region, it was not absolute. This supports the core hypothesis of the project—that health inequalities cannot be fully explained by economic or regional boundaries alone.

Some countries—like Sri Lanka, despite being classified as lower-middle income—appeared in more favorable clusters due to strong maternal health programs. Meanwhile, countries like Russia or some Gulf states, though wealthier, appeared in less favorable clusters due to high adult mortality or other non-communicable health burdens.

Sample Countries Per Cluster

```
🌐 Sample countries in Cluster 0:  
['Namibia', 'Eswatini', 'Malawi', 'Zambia', 'Lesotho']  
  
🌐 Sample countries in Cluster 1:  
['Lao PDR', 'Caribbean small states', 'Cambodia', 'Nepal', 'Timor-Leste']  
  
🌐 Sample countries in Cluster 2:  
['Africa Eastern and Southern', 'Pakistan', 'Angola', 'Sudan', 'Gambia, The']  
  
🌐 Sample countries in Cluster 3:  
['Estonia', 'Luxembourg', 'Ireland', 'United States', 'Slovenia']
```

Figure 15. Sample Countries in Each Cluster

Providing concrete examples of countries within each cluster helped ground the analysis in real-world context and illustrated how the clustering model mapped abstract health metrics to national realities.

Why This Approach Was Chosen

I chose to include PCA + K-means in this project because it represents a natural progression from regression analysis to unsupervised learning, enabling exploration without prior labels. It allowed me to detect latent structures in the data, and the results directly contribute to understanding how health inequalities manifest across multiple variables simultaneously—a key goal of the project. Moreover, clustering promotes data-driven categorization over externally imposed classifications, which is especially relevant in global health discourse where political, economic, and cultural narratives often shape perception more than empirical reality.

The results affirm that countries can be grouped by similar health system outcomes, independent of geography or GDP. This clustering provides a compelling case for rethinking how we frame global health interventions. Instead of targeting nations by income or location alone, policies could be more tailored to the specific health system profiles revealed through clustering. For example, countries in Cluster 2 could benefit from targeted maternal care investment and expanded vaccination programs, regardless of income classification.

3.5 3D Mesh Surface Plot

In this final visualization step, I sought to explore how multiple health indicators jointly influence life expectancy in a more visually dynamic and multidimensional way. This directly addresses the fifth research question: "How do multiple interacting health factors jointly influence life expectancy across countries?"

While previous visualizations revealed pairwise relationships and categorical differences, they often flattened complexity. To deepen the analysis, I constructed a 3D mesh surface plot showing how life expectancy varies across a two-dimensional predictor space—specifically, maternal mortality and health expenditure.

The decision to use a 3D mesh was both analytical and pedagogical. Traditional 2D plots fail to capture the interaction between two independent variables and their joint effect on an outcome. A 3D surface allows for a more holistic view of interaction effects, especially when non-linear relationships are present. This plot was also inspired by concepts from the computer graphics portion of the course—including mesh generation, lighting, and surface rendering. It blends data visualization with graphics principles to create a surface that not only displays data but communicates structure.

Building the Mesh Surface

3D Mesh: Life Expectancy vs Health Spending & Maternal Mortality

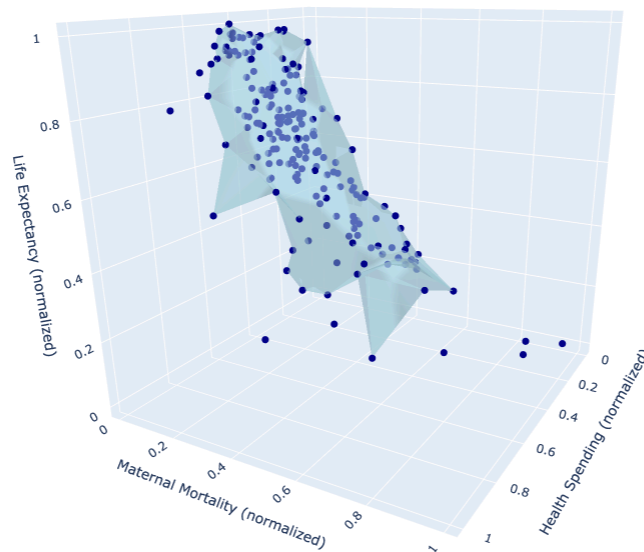


Figure 16. 3D Mesh Surface Plot - Life Expectancy vs. Health Spending & Maternal Mortality

To construct the mesh, I chose health expenditure (% of GDP) and maternal mortality ratio as the two predictor axes. These were selected based on previous analysis (correlation heatmap, regression results) where they showed strong and opposite effects on life expectancy. The response surface—life expectancy—was modeled by fitting a regression surface over the predictor grid using `scipy.interpolate.griddata`. The result is a smooth, continuous surface showing how life expectancy changes across the two-variable space.

The 3D plot was implemented using **Plotly**, chosen for its ease of 3D rendering and smooth interactivity. Though ultimately presented as a static image in the PDF report, the interactive version allows users to rotate, zoom, and explore contours—making it a powerful exploratory tool.

Interpreting the Visualization

The surface reveals a clear gradient—high health expenditure and low maternal mortality are associated with high life expectancy, forming a plateau at one corner of the mesh, low expenditure and high maternal mortality correspond to a sharp decline in life expectancy, forming a deep valley in the opposite corner.

What stands out is the non-linear curvature of the surface, which suggests diminishing returns on health spending in countries where maternal mortality remains high. In other words, increasing expenditure alone does not ensure better outcomes unless it is targeted toward critical care areas like maternal health.

This visualization reinforces prior findings from regression and clustering: systemic weaknesses in maternal and child health services have a disproportionately large impact on national health outcomes.

From a technical standpoint, this mesh plot echoes the surface rendering principles covered in CG labs, where 3D objects are defined by vertex grids and color-mapped surfaces. Concepts like vertex

interpolation, shading, and mesh topology translate naturally to this data-driven context. While not a real-time 3D model, the construction of this surface involved many of the same geometric and visual design principles. It stands as a demonstration of how computer graphics techniques can enrich data storytelling, especially in multivariate domains.

4. Discussion

The multivariate approach adopted in this project reveals a series of rich, interlinked insights that collectively respond to the guiding research questions. Each visual and analytical method contributed a distinct perspective, and their integration created a holistic view of global health disparities.

Which indicators are most predictive?

Across correlation heatmaps, OLS regression, and the 3D surface plot, the analysis consistently showed that maternal mortality, infant mortality, and fertility rate are among the most powerful negative predictors of life expectancy. Health expenditure, while positively correlated, revealed diminishing returns in certain contexts—especially when maternal health outcomes remained poor. These findings emphasize that investment in health must be strategic, not just abundant. Preventive care and maternal/child health services emerge as particularly impactful areas.

How does inequality manifest regionally?

Regional disparity was evident in both visual and statistical outputs. Sub-Saharan Africa and parts of South Asia consistently appeared at the lower end of health outcomes, regardless of whether the analysis was spatial (choropleth maps), temporal (line plots), or statistical (t-tests). However, clustering analysis revealed that these regional patterns do not fully define a country's health profile—some nations deviated from their regional norm due to strong public health infrastructure or other unique factors.

Are country profiles clusterable?

Yes. PCA and K-means clustering showed that countries could be grouped into meaningful clusters based on multiple health indicators. These clusters did not map perfectly onto income groups or geographic regions, reaffirming the value of data-driven classification. Importantly, this highlighted cases where countries outperformed or underperformed relative to their economic status—e.g., Sri Lanka outperforming its income group due to robust maternal health interventions.

How do combined factors shape life expectancy?

The 3D mesh plot served as a compelling culmination, illustrating how life expectancy results from the interaction of multiple health factors. By visualizing the nonlinear interplay between maternal mortality and health expenditure, the surface plot conveyed a nuanced message: money alone is not enough. Reducing maternal mortality unlocks greater returns on investment. This analysis visually and analytically linked previously isolated patterns, providing an intuitive yet powerful message about policy prioritization.

Strengths of visualisation types

Each visual technique was chosen purposefully. Bar plots and line charts provided familiarity and clarity, heatmaps offered compact overviews of variable interactions, PCA and clustering extended the analysis into pattern discovery, and the 3D mesh integrated computer graphics concepts to explore multivariate interaction. This diversity not only met the technical goals of the course but also created

an engaging, layered narrative. Together, they showcase the value of visual storytelling in data science.

5. Limitations

While this project uncovers meaningful insights, it also carries limitations that must be acknowledged.

Data sparsity and regional imbalance

Some countries, especially smaller or conflict-affected regions, had missing or inconsistent data over time. Even after forward/backward imputation, this introduces uncertainty, particularly in time-series analyses or cluster assignments.

Static country-level analysis

The project used country-year data but aggregated or imputed across time for clarity and consistency. This sacrifices some temporal detail and may obscure within-country changes. Moreover, the analysis is observational; while correlations and regression coefficients are informative, they do not imply causation.

Clustering depends on selected features

The PCA and K-means results are contingent on the choice and normalization of variables. Different sets of indicators or feature transformations might produce different clusters. Additionally, K-means assumes convex cluster shapes and equal variances, which may not fully capture the true structure of global health data.

Despite these limitations, the insights remain valuable within the project's exploratory scope. They point toward areas for further research, including more dynamic and causal modeling.

6. Conclusion and Reflection

This project set out to explore global health inequality by using data analysis and visualisation techniques to understand how life expectancy and related health outcomes differ across countries. By using a wide range of tools—from simple bar charts and correlation maps to clustering and 3D surface plots—it was possible to uncover important patterns and relationships in the data. Each visualisation played a role in building a bigger picture, helping to answer key questions about where and why health outcomes vary, and what factors matter most.

One clear finding from this project is the crucial role of maternal and child health. Countries with higher rates of maternal and infant mortality, along with higher fertility, tend to have much lower life expectancy. This pattern was consistent across different methods of analysis, including regression and clustering. It shows that improving healthcare for mothers and children could lead to big gains in overall health, even in countries with limited resources. The clustering results also showed that countries can be grouped based on their health profiles in ways that don't always match income level or geographic region—suggesting that tailored, data-driven policies may be more effective than one-size-fits-all solutions.

While there were challenges—such as missing data and the limits of drawing strong conclusions from correlations—the project shows how different techniques can be combined to explore complex problems. The visualisations were chosen to suit the types of questions being asked, and each stage

built on the last to give a fuller understanding. Through this process, I was able to apply and extend many of the skills I've learned throughout the course, from basic plotting to more advanced methods like PCA and mesh-based graphics. This project brought everything together in a way that felt meaningful and rewarding.

On a personal note, this topic was especially important to me. Coming from a country where access to healthcare isn't always equal, I've seen firsthand how these issues affect people's lives. Choosing this topic allowed me to connect what I've learned in class with real-world challenges I care about. That personal connection helped keep me motivated, even when the project got complex or time-consuming.

In the end, this project shows how data visualisation—when combined with clear analysis and thoughtful storytelling—can help us understand big global problems in a more human and meaningful way. I've not only gained technical skills but also learned how to communicate insights in a way that's engaging and informative. This has been a challenging, creative, and deeply worthwhile experience, and I'm proud of how it all came together.

References

- Qurban, B. (2023). *World Health Indicators Dataset*. Kaggle. <https://www.kaggle.com/datasets/bushraqurban/world-health-indicators-dataset/>
- Portions of this report were drafted with the assistance of generative AI (ChatGPT) and Grammarly to help structure explanations and improve clarity.